MATH 241: Calculus III

Fall 2026

Lecture 6: Functions of Several Variables

Background & Motivation

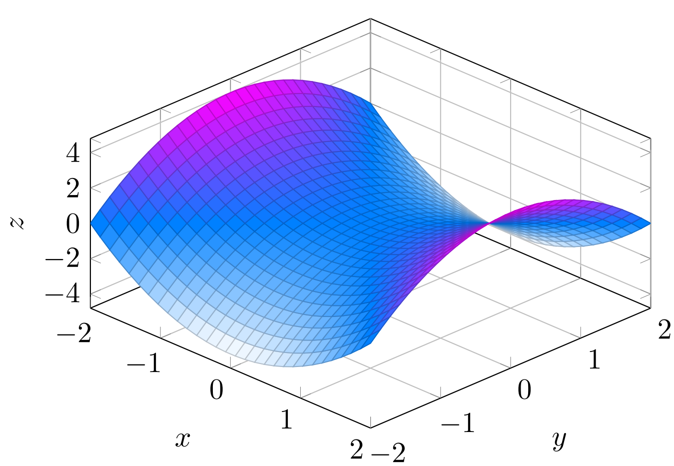
This is where Calculus III truly begins. Until now we've worked with vectors and curves—extensions of familiar geometry. Starting today, we study functions $f(x, y)$ and $f(x, y, z)$ that take multiple inputs and return a single output. Temperature at a point in a room, pressure in the atmosphere, cost as a function of two inputs—these are all multivariate functions. We need new tools to visualize them (level curves, cross-sections) and new techniques to analyze them (partial derivatives, gradients, multiple integrals). Everything from here to the end of the course builds on this foundation.

1. Functions of Two Variables

Definition 1: Function of Two Variables

A **function of two variables** $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ assigns to every point (x, y) in its domain a single real number $f(x, y)$. Whereas a single-variable function $y = f(x)$ draws a curve in the plane, a two-variable function draws a **surface** in 3D: for each input (x, y) you plot the point $(x, y, f(x, y))$, and the set of all such points is the **graph** $z = f(x, y)$. Think of the (x, y)-plane as the "floor" and the surface as a landscape whose height above that floor is determined by f.
– The graph $z = f(x, y)$ is a **surface in** $\mathbb{R}^3$

Surface Plot of $z = x^2 - y^2$



Graph of $z = f(x, y)$ as a 3D surface. Each point (x, y) in the plane is lifted to height $f(x, y)$; here $z = x^2 - y^2$ is the classic saddle.

Example 1: Real-World Multivariable Functions

Write each as $f(\text{inputs})$:

(a) Ohm's Law: $V = IR$ (V depends on current I and resistance R)

(b) Kinetic energy: $KE = \frac{1}{2}mv^2$ (depends on mass m and velocity v)

(c) Ideal gas law: $P = \frac{nRT}{V}$ (at fixed n, R: depends on T and V)

2. Domains and Ranges

Domain Restrictions to Watch For

- **Denominators** – cannot equal 0
- **Square roots** – argument ≥ 0
- **Logarithms** – argument > 0
- **Inverse trig** – $|\text{argument}| \leq 1$ (for arcsin, arccos)

Example 2: Finding Domains

Find the domain of each function:

(a) $f(x, y) = \dfrac{1}{x + y}$

1. Restriction: $x + y \neq 0$

2. Result: $\text{dom}(f) = \{(x, y) : y \neq -x\}$ – all of $\mathbb{R}^2$ except the line $y = -x$

Range: $(-\infty, 0) \cup (0, \infty)$

(b) $g(x, y) = \dfrac{\ln(x - y)}{\sqrt{x^2 + y^2 - 4}}$

1. Restrictions: $x - y > 0$ (for ln) AND $x^2 + y^2 - 4 > 0$ (for $\sqrt{\ }$ in denominator)

2. Result: $\text{dom}(g) = \{(x, y) : x > y \text{ and } x^2 + y^2 > 4\}$

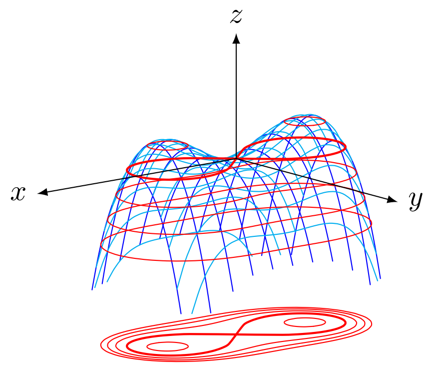
– Region below the line $y = x$ and outside the circle of radius 2

3. Level Curves

Definition 2: Level Curves

A **level curve** (or **contour**) of $f(x, y)$ at height k is the set of all points in the (x, y)-plane where f takes the value k — equivalently, the horizontal slice of the graph at height $z = k$, projected down onto the floor. Drawing many of these for different values of k produces a **topographic map** of the surface: regions where curves are packed close together correspond to *steep* parts of the surface, while widely-spaced curves correspond to gentle slopes.

$$\{(x, y) : f(x, y) = k\}$$



A 3D surface with its level curves projected straight down onto the xy-plane. The infinity-shaped level curve is a Bernoulli lemniscate. Together, the surface and its contour projection give complementary views of the same function.

Plotting Level Curves

1. Choose several values of k in the range of f
2. For each k, set $f(x, y) = k$ and simplify
3. Identify and sketch each resulting 2D curve
4. Label each curve with its k-value

Example 3: Level Curves of a Paraboloid

Sketch level curves of $f(x, y) = x^2 + y^2$ for $k = 1, 4, 9, 16$.
1. Set up: $x^2 + y^2 = k$ – circles of radius $\sqrt{k}$
2. Identify curves:

- $k = 1$: circle of radius 1
- $k = 4$: circle of radius 2
- $k = 9$: circle of radius 3
- $k = 16$: circle of radius 4

3. Result: Concentric circles – they get closer together as k increases (steeper slope at higher z).

Example 4: Level Curves of a Saddle

Sketch level curves of $f(x, y) = x^2 - y^2$ for $k = -4, -1, 0, 1, 4$.
1. Set up: $x^2 - y^2 = k$
2. Identify curves:

- $k > 0$: hyperbolas opening along the x-axis
- $k < 0$: hyperbolas opening along the y-axis
- $k = 0$: $y = \pm x$ (two lines through origin)

3. Result: Topographic map of a hyperbolic paraboloid (saddle surface). Two families of hyperbolas cross at the saddle point.

4. Common 3D Equations

Surfaces You Should Recognize

Equation	Surface
$x^2 + y^2 + z^2 = r^2$	Sphere
$x^2 + y^2 = r^2$	Cylinder
$z = x^2 + y^2$	Paraboloid
$x^2 + y^2 = z^2$	Cone
$z = x^2 - y^2$	Saddle (hyperbolic paraboloid)
$Ax + By + Cz = D$	Plane

5. Functions of Three or More Variables

Definition 3: Level Surfaces

For $f : \mathbb{R}^3 \to \mathbb{R}$, we cannot graph $w = f(x, y, z)$ (requires 4D). Instead we use **level surfaces**:

$$\{(x, y, z) : f(x, y, z) = k\}$$

– The 3D analog of level curves

Example 5: Level Surfaces of $x^2 + y^2 + z^2$

Describe the level surfaces of $f(x, y, z) = x^2 + y^2 + z^2$.

1. Set up: $x^2 + y^2 + z^2 = k$

2. Identify:

- $k > 0$: spheres of radius $\sqrt{k}$ centered at the origin
- $k = 0$: single point (the origin)
- $k < 0$: empty set

3. Result: f measures squared distance from origin, so level surfaces are concentric spheres.

Practice Problems

Problem 1. Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

Answer: $x^2 + y^2 \leq 9$ – the closed disk of radius 3

Problem 2. Find the domain and range of $f(x, y) = e^{-(x^2+y^2)}$.

Domain: all of $\mathbb{R}^2$ (exponent always defined).

Range: $(0, 1]$. Max $f = 1$ at origin; $f \to 0$ as $\|(x, y)\| \to \infty$.

Problem 3. Sketch level curves of $f(x, y) = xy$ for $k = -2, -1, 0, 1, 2$.

$xy = k \implies y = k/x$: hyperbolas with asymptotes on the axes.

$k = 0$: the axes themselves ($x = 0$ or $y = 0$).

$k > 0$: curves in Q1 and Q3. $\quad k < 0$: curves in Q2 and Q4.

Problem 4. Describe the level surfaces of $T(x, y, z) = 100 - x^2 - y^2 - z^2$.

$T = k \implies x^2 + y^2 + z^2 = 100 - k$. Spheres of radius $\sqrt{100 - k}$ centered at origin.

Valid for $k \leq 100$. Max temperature $T = 100$ at origin.

Problem 5. The wind chill index is $W(v, T) = 35.74 + 0.6215T - 35.75v^{0.16} + 0.4275Tv^{0.16}$. Find W at $T = 0°\text{F}$ and $v = 25$ mph. What do the level curves represent physically?

$v^{0.16} = 25^{0.16} \approx 1.673$

$W = 35.74 + 0 - 35.75(1.673) + 0 = 35.74 - 59.83 \approx -24.1°\text{F}$

Level curves $W = k$: combinations of wind speed and temperature that "feel" the same.